After the credits

Personalized recommendations with limited rating history

Project plan | MAT 422 | Aashir Javed | October 25, 2026

1 Introduction

I consider myself a cinemaholic. I've watched thousands of movies and have an AMC Stubs A-List subscription, and I usually watch three or four movies each week. Math and movies are two things I love, so I am especially interested in the mathematics behind personalized recommendations.

A recommendation service has little to work with when someone has rated only a few movies. I want to find out when those ratings become enough to improve on a popular-movies list. This project asks whether recommendations improve as we reveal 5, 10, or 20 ratings from the same person, and whether adding genre preferences helps when the history is short. The goal is to compare how much useful personalization each approach provides with limited information.

Two hypotheses will guide the comparison: with five ratings, a hybrid of genre information and matrix factorization will have higher mean NDCG@10 than matrix factorization alone; with twenty ratings, matrix factorization will outperform the popularity baseline on the same metric. These are outcomes to test, not results already established.

2 Related work

MovieLens provides an established source of explicit movie ratings [1]. Koren, Bell, and Volinsky explain how matrix factorization represents users and movies with short vectors and accounts for rating biases [2]. Burke describes ways to combine recommendation methods and evaluates a hybrid system [3], which motivates checking whether genres add useful information here. Krichene and Rendle show that sampling candidate items can distort ranking comparisons [4], so the evaluation will rank the full eligible catalog. These studies do not settle which of our four approaches works best under the same 5-, 10-, and 20-rating limits. This project will address that specific comparison using established methods.

3 Proposed methodology

The popularity baseline will rank movies by the number of four- or five-star ratings in the training data. A genre model will represent each movie as a binary genre vector. A user's profile will combine those vectors, weighting each revealed rating by its difference from three stars. Cosine similarity will score candidate movies; an empty or zero-length profile will fall back to popularity.

Low-rank matrix factorization will learn short user and movie vectors, a global rating mean, and user and movie biases. Predicted ratings combine the mean, biases, and the vectors' dot product. Alternating regularized least squares will fit the model using observed ratings only. For an evaluation user, the learned movie vectors stay fixed while a regularized least-squares fit estimates that user's vector and bias from the revealed ratings. Missing ratings will remain unknown.

The hybrid will average the genre and factorization scores after converting each to percentile ranks within the candidate catalog, with a weight chosen on validation users. I will compare 10, 20, and 40 factors, regularization strengths of 0.1, 1, and 10, and hybrid weights of 0, 0.25, 0.5, 0.75, and 1. Python notebooks using NumPy, pandas, SciPy, and Matplotlib will keep the matrix operations and evaluation easy to follow.

4 Experiment setups

The data will come from the fixed MovieLens 1M release downloaded from GroupLens [1]. It contains 1,000,209 ratings from 6,040 users. The files provide user and movie IDs, one- to five-star ratings, timestamps, titles, and genres. Demographic fields will not be used. Code, package versions, split IDs, and data-access instructions will be saved on GitHub; the raw dataset will not be redistributed.

A fixed seed of 422 will split user IDs into 70% background training, 15% validation, and 15% final testing. Only background users will train the shared movie factors and popularity counts. Validation and test users must have at least 30 ratings for movies present in that training catalog. Their earliest 20 recorded ratings will form a profile pool; all remaining ratings will be held out. Revealing the first 5, 10, or 20 pool ratings keeps the evaluation ratings fixed. All 20 pool movies will be excluded from recommendations in every condition, keeping candidate sets consistent. This is a controlled offline study of limited histories, not a simulation of a live release timeline.

Settings will be selected by mean NDCG@10 on validation users, then frozen before scoring final test users. NDCG@10 will measure how highly withheld positive ratings appear; Recall@10 will measure the fraction recovered. Four- and five-star ratings count as positive. Ranking averages will include users with at least one eligible held-out positive, and exclusion counts will be reported. Catalog coverage will measure the fraction of eligible movies recommended across those users. Paired bootstrap resampling of users will provide 95% confidence intervals for the two hypothesis comparisons using 1,000 resamples. A performance-versus-history plot and coverage table will summarize the results.

5 Expected results

I expect genre information to be most helpful with short histories, while matrix factorization should benefit from additional ratings. The comparison may instead show that a simple baseline remains competitive. Either outcome will help explain when the extra modeling is useful. The older catalog, coarse genres, and focus on users with enough ratings limit generalization. Unrated movies are unjudged, so offline scores cannot establish whether someone would enjoy every recommendation.

Individual responsibility. This is a solo project. Aashir Javed is responsible for data preparation, implementation, evaluation, and the final report, and is the lead writer for all five sections.

AI assistance. I came up with the movie recommendation topic, proposed, and made the initial draft. Codex helped with background research of stuff like datasets etc, strengthening the proposal and final wording changes. I confirm that i have reviewed and checked the sources, code, and results and documentiation.

References

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